

Christy Cheung

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EDUCATION

Cornell University, Ithaca, NY

GPA: 3.96/4.0

Bachelor of Science in Computer Science, Minor in AI

Relevant Courses: Object-Oriented Programming, Data Structures, Machine Learning, Computer Vision

EXPERIENCE

VortexNet

Dec. 2025 – Jan. 2026

Software Engineering Intern

- Developed an **end to end** audio-to-text **web application** using **OpenAI models** to streamline addition of audio transcriptions and video subtitles for 50+ languages in collaboration with product and ML teammates
- Implemented **JWT-secured REST APIs** with FastAPI and **Celery + Redis async job orchestration** and in-memory **caches** to offload ML inference from API services, maintaining P90 request latency under 500 ms
- Delivered a responsive **user interface using React** with client-side routing and type-safe **TypeScript SDK**, ensuring consistent frontend-backend API contracts, reducing integration bugs across 6 REST endpoints + 5 frontend routes
- Migrated from in-memory storage to **PostgreSQL relational databases** with SQLAlchemy and Alembic, eliminating data loss on restarts and ensuring transactional consistency across API services and background workers
- Deployed a containerized backend API on **AWS ECS** behind an **Application Load Balancer**, integrating **RDS**, **ElastiCache**, and **S3** for scalable cloud deployment and team-wide **CI/CD workflows**

KPMG

July 2025 – Aug. 2025

Machine Learning Intern

- Partnered with multidisciplinary compliance teams to **automate sensitive information detection workflows across 3 enterprise applications**, reducing manual compliance review time by 60.3%
- Customized object detection models and automation scripts to recognize 15+ sensitive stamp types on financial documents, achieving 93% detection accuracy and improving audit reliability
- Spearheaded adoption of **computer vision models** in cross-functional teams, reducing compute and memory usage and improving efficiency of large-scale document processing pipelines

EmPRISE Lab

Apr. 2025 – Jan. 2026

Undergraduate Research Assistant

- Fine-tuned YOLO **pose estimation models** and built **data processing pipelines for patient care**, eliminating 7,600 hours of manual labeling and improving model precision by 80.9% in a collaborative research setting
- Optimized **3D pose reconstruction** pipeline using genetic algorithms and NumPy, processing 162 scenes, improving accuracy by 40.5%, **creating data visualization modules** for testing and **detailed documentations**
- Published multi-modal dataset for assistive robots** at the IEEE/RSJ IROS and **benchmark for LLM care-giving decision-making** abilities in ACM/IEEE HRI, contributing 2 resources for healthcare robotics research

PROJECTS

Physical Caregiving Language Model

Sep. 2025 – Feb. 2026

- Devised an AI workflow to **retrieve patient-specific clinical evidence** from 270,000+ medical publications, supporting user functionality estimation and downstream robot caregiving policy
- Improved response accuracy by **37%** on benchmark tasks by integrating a **Retrieval-Augmented Generation (RAG)** pipeline using LangChain and **Claude Code** with in-context learning and chain-of-thought prompting
- Collaborated with clinical partners** to translate patient evaluation into language model reasoning frameworks
- Automated dataset curation from 5 million+ articles with **OpenAI APIs**, reducing manual processing time by 80%

Experiential Robotics (XRP) Projects

Nov. 2024 – Present

- Built a line-following robot using sensor fusion and PID control, and deployed it on an **open-source robotics platform** to support hands-on instruction for students
- Coordinated with Boeing engineers to develop an interactive International Space Station model, implementing software to mirror real-time telemetry for educational and outreach use
- Created CAD and optimized humanoid arms to increase load capacity by 50% and reduce assembly time by 77%

TECHNICAL SKILLS

Programming: Python, Java, TypeScript, SQL, HTML/CSS, OCaml

Frameworks and Tools: React, FastAPI, Redis, Celery, Material UI, SQLAlchemy, Alembic, scikit-learn

Cloud and DevOps: AWS (ECS, ECR, RDS, ElastiCache, CloudWatch, IAM), Docker, GitHub Actions, Git